Fields and Waves I

Lecture 4

Transient Analysis of Transmission Lines

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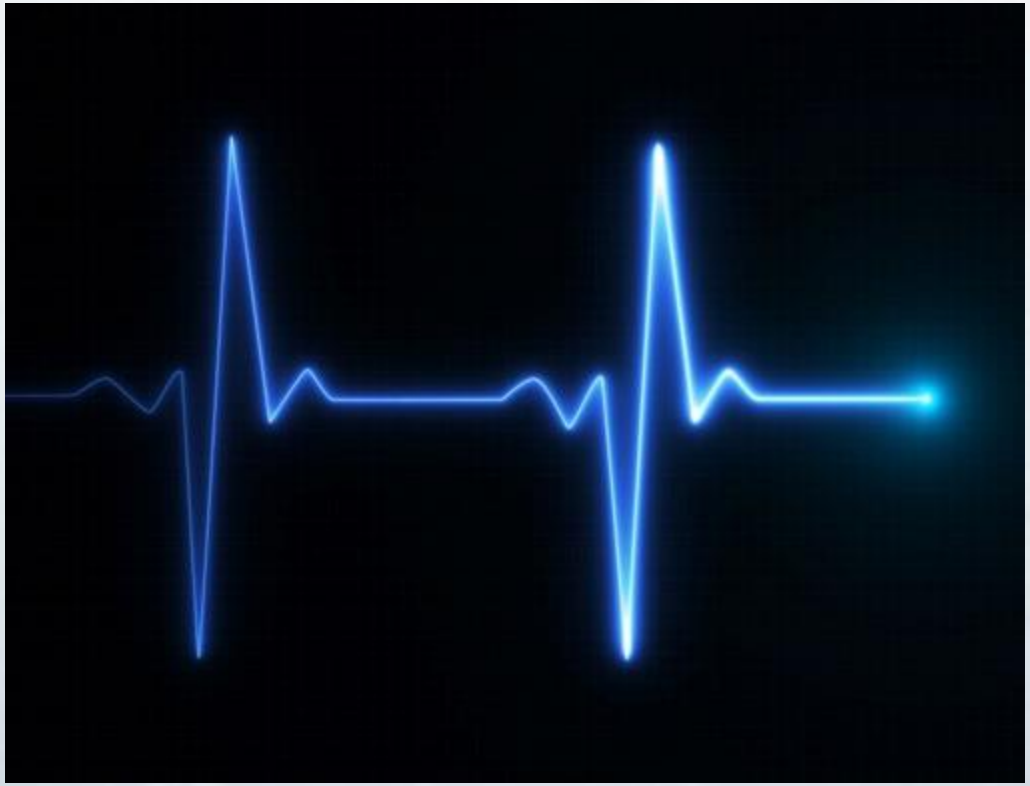
These slides were prepared through the work of the following people:

- Kenneth A. Connor – ECSE Department, Rensselaer Polytechnic Institute, Troy, NY (Emeritus)
- J. Darryl Michael – GE Global Research Center, Niskayuna, NY
- Thomas P. Crowley – National Institute of Standards and Technology, Boulder, CO
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- Jeffrey Braunstein – ECE Department, University at Albany
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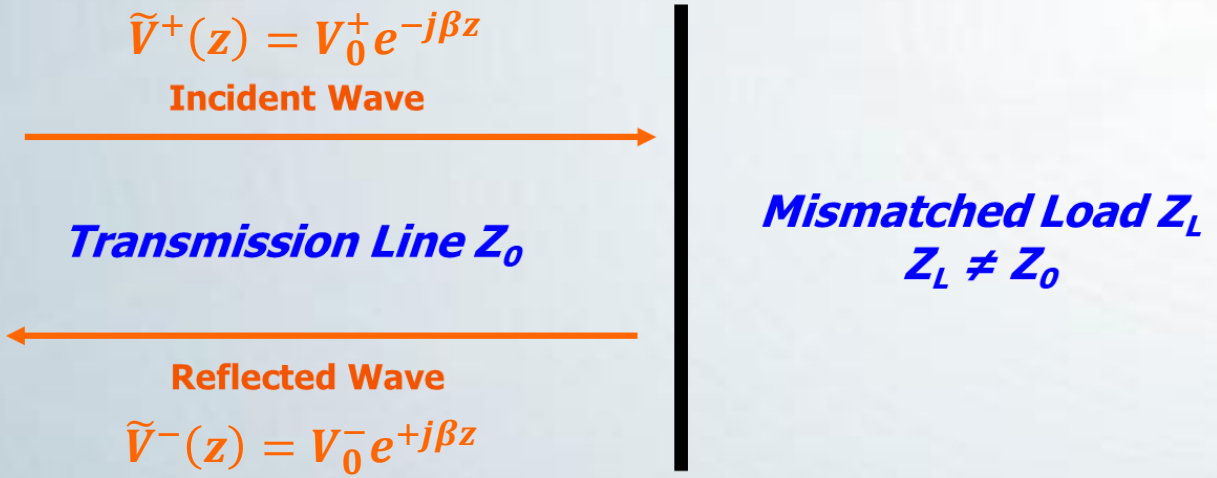
Materials from other sources are referenced where they are used.
Those listed as Ulaby are figures from Ulaby's textbook.

Outline

- Transients On Transmission Lines
- Bounce Diagrams



Review



- How much gets reflected?

$$V_0^- = \Gamma V_0^+ \longrightarrow \Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

Reflection Coefficient

Review

- If there is reflection at the load, the total voltage on the t-line is now the sum of the incident and reflected waves

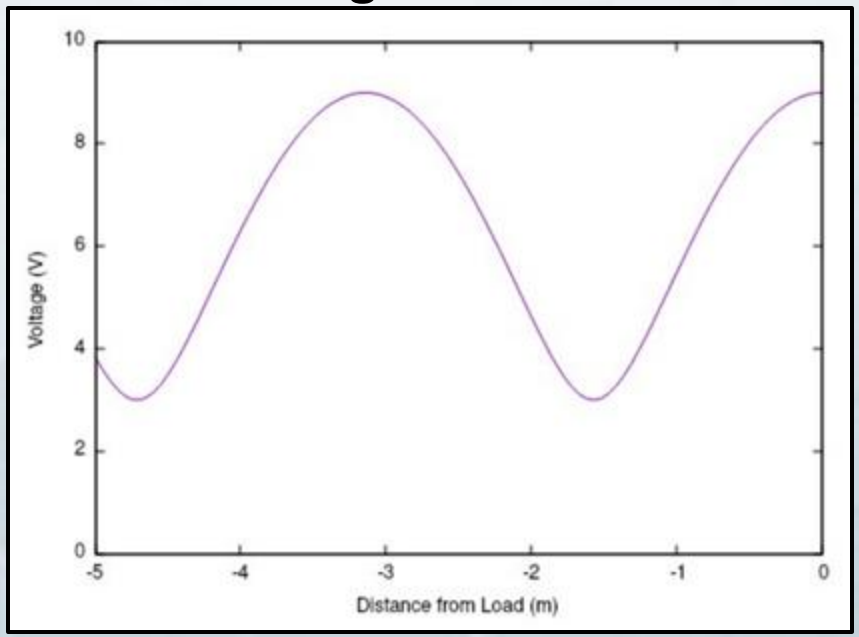
$$\tilde{V}(z) = V_0^+ e^{-j\beta z} + \Gamma V_0^+ e^{+j\beta z}$$

- This means that the voltage wave is no longer a pure traveling wave, but is partially a **standing wave** as well

- If we are given a **standing wave pattern**, we can determine several useful quantities
  - Voltage wavelength
  - SWR (standing wave ratio)
  - Reflection coefficient
  - Load impedance

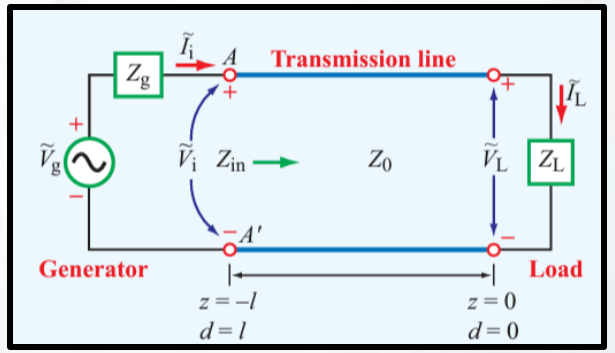
$$\textbf{SWR} = \frac{V_{max}}{V_{min}} = \frac{1+|\Gamma_L|}{1-|\Gamma_L|}$$

Standing Wave Pattern



Review

- If the ratio of the voltage to the current gives an **impedance**, we can define an impedance based on the voltage and current at a certain point on the transmission line $z = -d$:



$$Z_d = \frac{\tilde{V}(d)}{\tilde{I}(d)}$$

- If we evaluate this ratio at the load (z=0): where Z_L is the load impedance at the end of the t-line

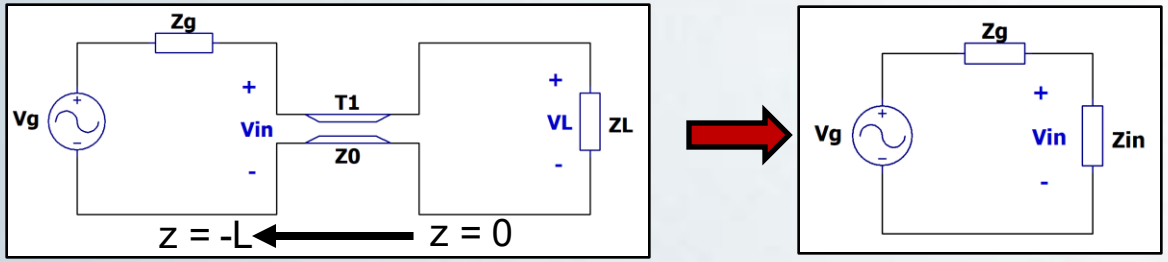
$$\rightarrow \quad Z_L = \frac{\tilde{V}_L}{\tilde{I}_L}$$

- If we evaluate this ratio at the input to the transmission line (z=-L), we get the **input impedance** of the t-line + load

$$\rightarrow \quad Z_{in} = \frac{\tilde{V}_{in}}{\tilde{I}_{in}}$$

Review

- In a circuit containing a t-line + load, we can replace the t-line + load with $\boldsymbol{Z_{in}}$:



- The full expression for $\boldsymbol{Z_{in}}$:

$$Z_{in}(L) = Z_0 \frac{Z_L + jZ_0 \tan(\beta L)}{Z_0 + jZ_L \tan(\beta L)}$$

- For an open-circuit load:

$$Z_{in}^{oc}(L) = -jZ_0 \cot(\beta L)$$

- For a short-circuit load:

$$Z_{in}^{sc}(L) = jZ_0 \tan(\beta L)$$

- For a line of length $L = \frac{n\lambda}{2}$:

$$Z_{in}^{\lambda/2}(L) = Z_L$$

- For a line of length $L = \frac{n\lambda}{4}$:

$$Z_{in}^{\lambda/4}(L) = Z_0^2/Z_L$$

Transforms load impedance to look like a different impedance

Review

The transmission line impedance transformer has some limitations:

- Input signal needs to be one frequency (different impedance for different frequencies)
- Only works at steady state (calculated Z_{in} is a steady state result only) – does not work with transient signals

To better understand transmission lines, we need to broaden our understanding to include **transient behavior**.

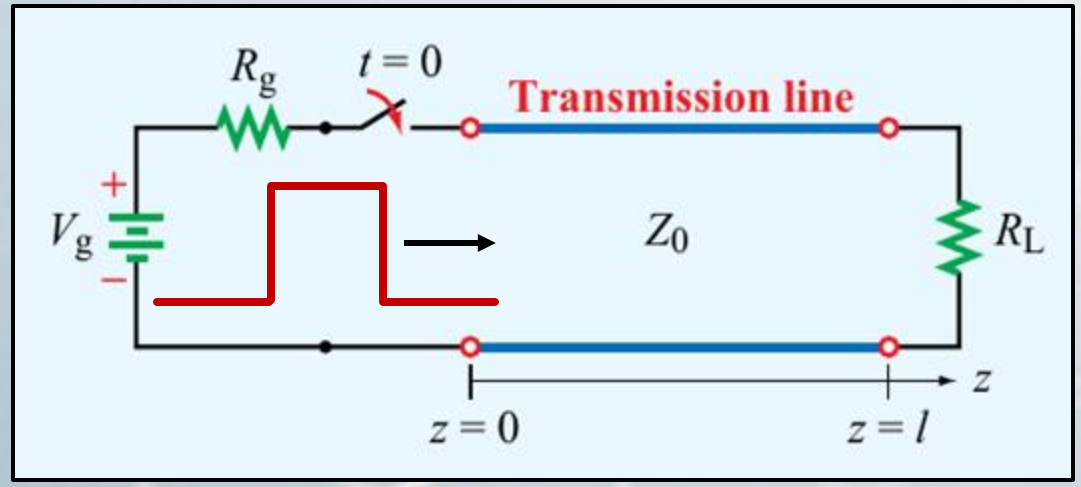
Transients on Lossless Transmission Lines

Transients On T-Lines

What do transient signals do on a lossless t-line?

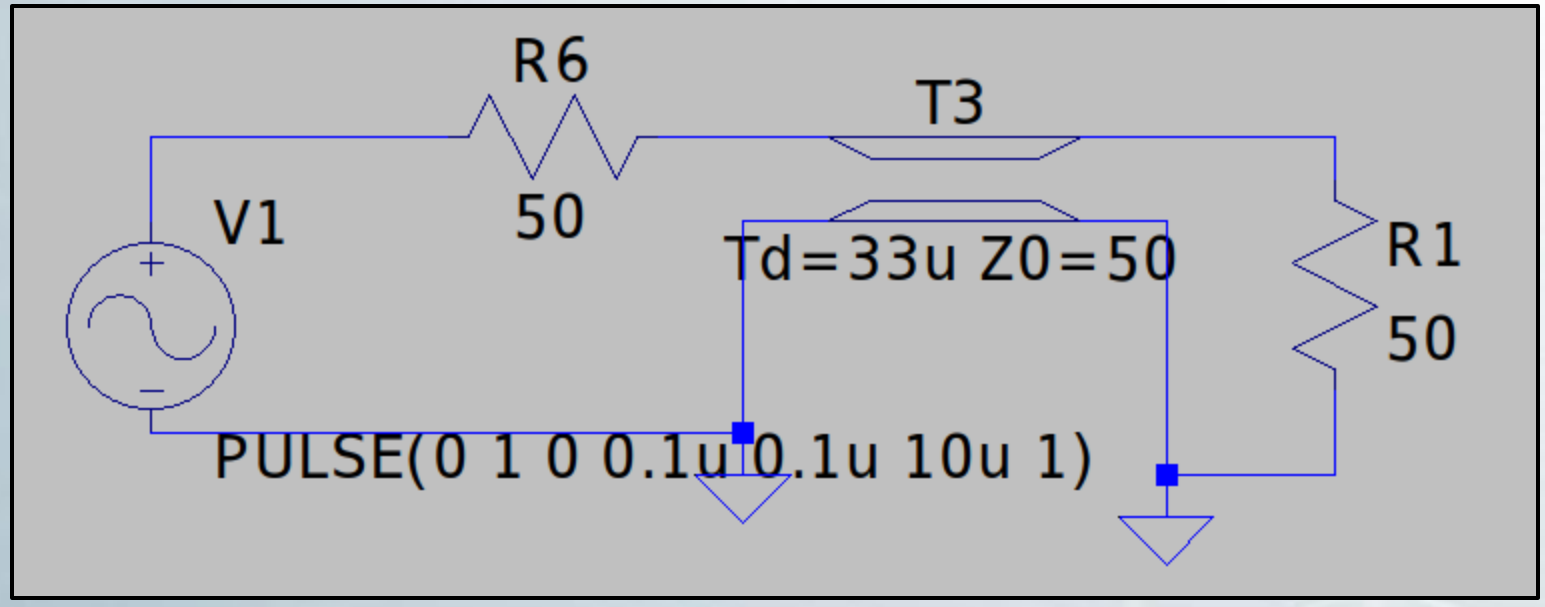
- Pulses will look the same at the t-line input and output except for a delay (no dispersion/distortion)
- This is independent of the shape of the pulse.

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Transients On T-Lines

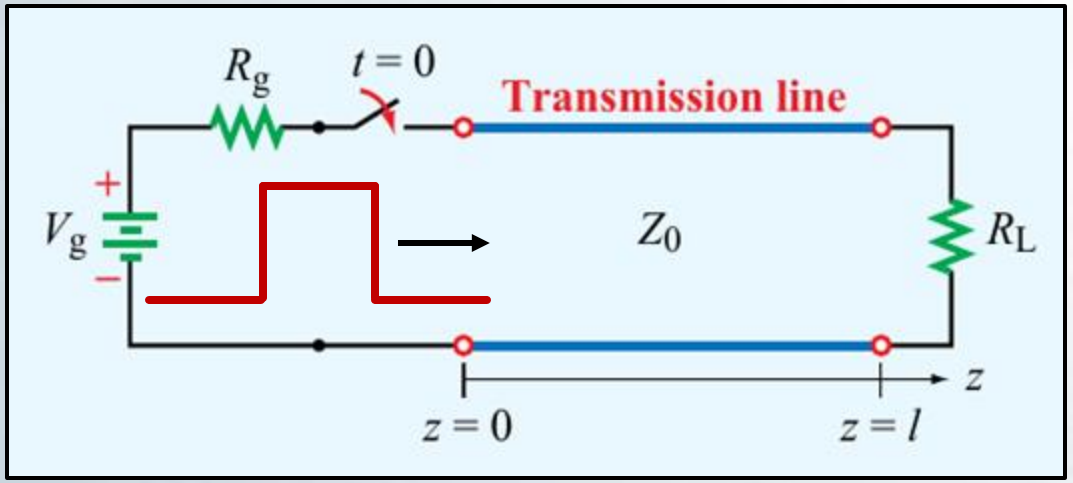
Basic simulated transmission line with pulsed source



How do we understand the voltage drop across R6?

Transients On T-Lines

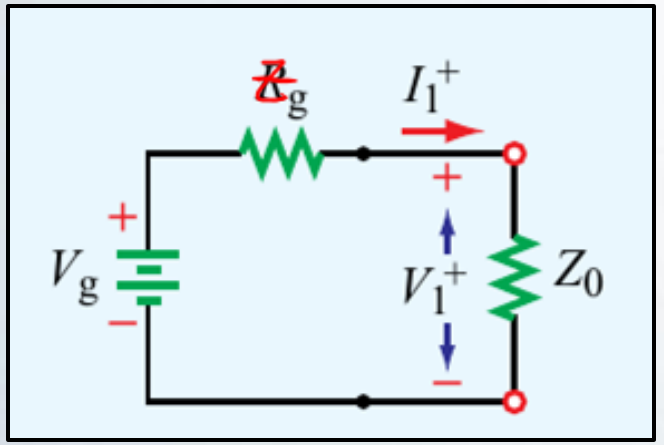
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Consider the circuit above. At t=0, the switch closes and a **voltage pulse** begins moving forward down the line.

- Is there a backwards traveling wave at this moment (t = 0^+)?
- Are there any standing waves?
- What is the ratio of voltage to current on the line at this moment?

Transients On T-Lines



Ulaby pg. 115

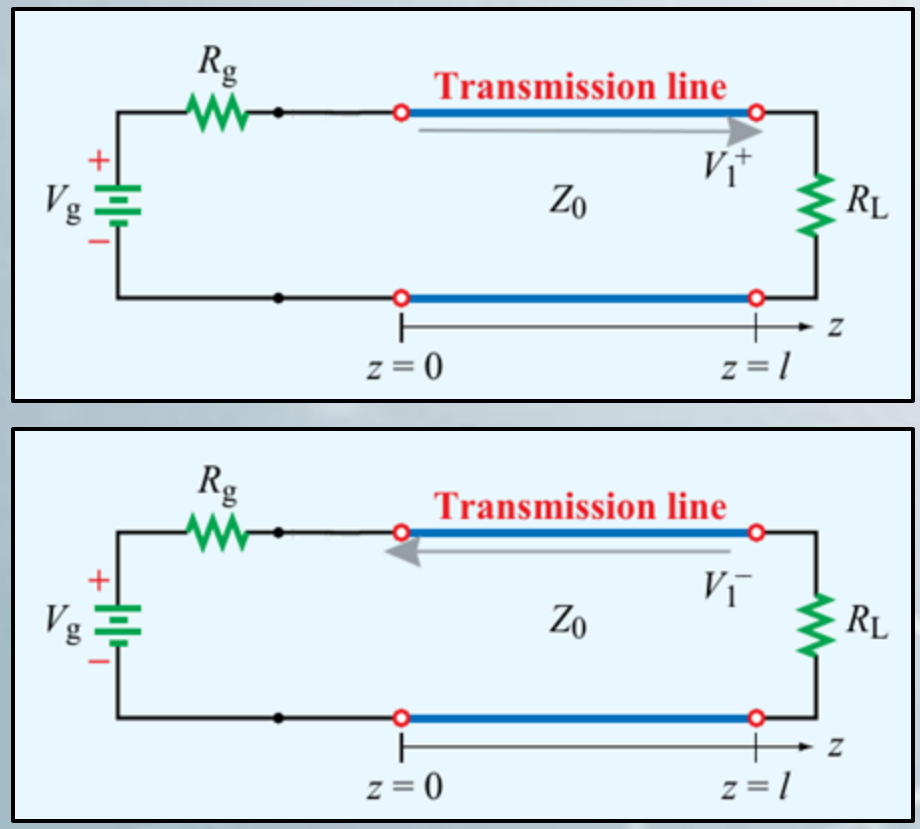
- At t=0^+, the t-line circuit therefore behaves equivalently to this circuit and the voltage and current on the t-line are:

$$V_1^+ = \frac{V_g Z_0}{Z_g + Z_0} \qquad I_1^+ = \frac{V_g}{Z_g + Z_0}$$

- But what are the current and voltage expressions at the load?

Transients On T-Lines

- When it reaches the load, the voltage pulse still reflects at the load as governed by the reflection coefficient.



1st reflected pulse amplitude

$$V_1^- = \Gamma_L V_1^+$$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$\Gamma_L = \frac{V_1^-}{V_1^+} = -\frac{I_1^-}{I_1^+}$$

Transients On T-Lines





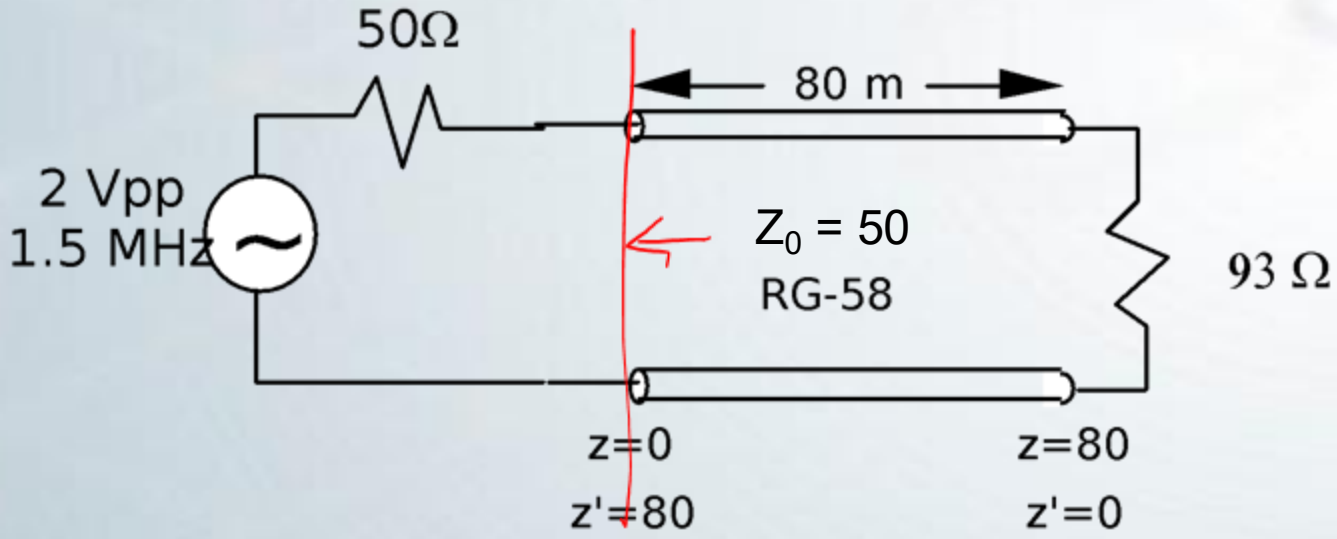
The pulse travels down the line

hits the load, reflects if there is an impedance mismatch



the reflection goes back toward the source.

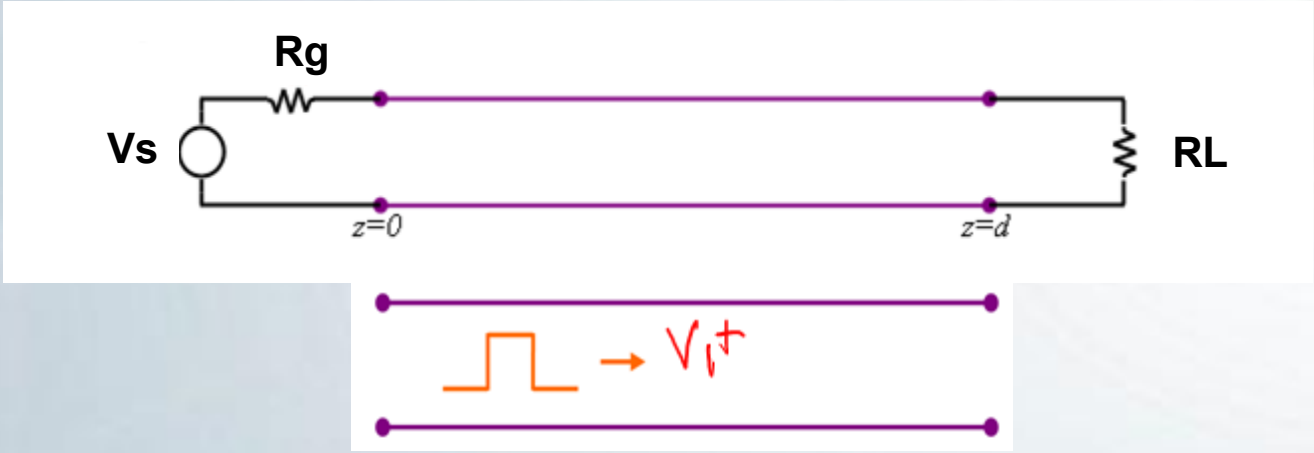
Transients On T-Lines



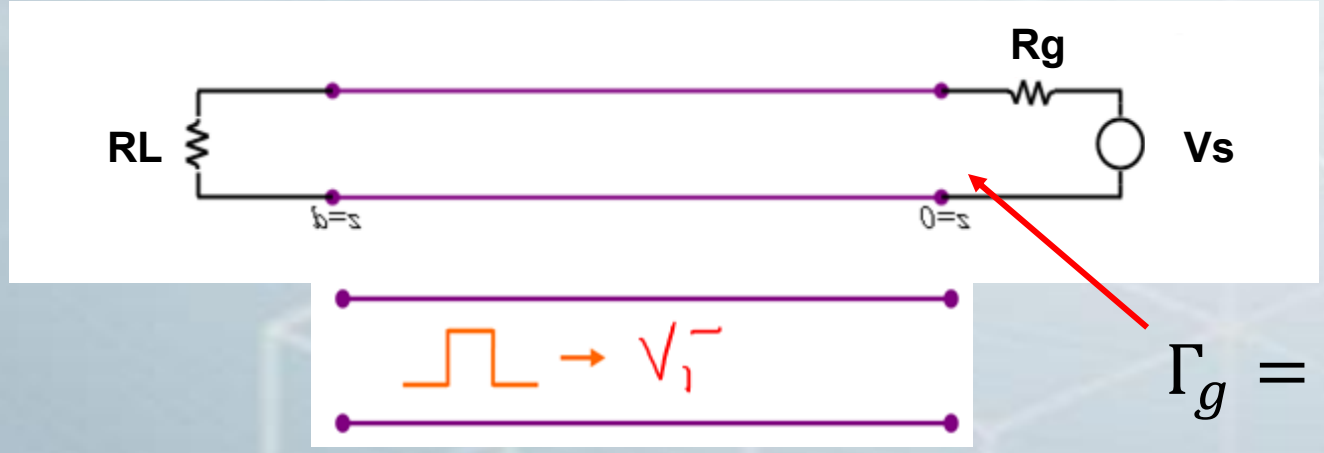
- The model t-line circuits we have considered thus far have had a source impedance, and that source impedance has tended to be equal to the characteristic impedance of the line.

Why is this?

Transients On T-Lines



Mathematically, there is no difference between a signal hitting a load impedance and a signal hitting a source impedance. In both cases, the behavior will be governed by the reflection coefficient.



$$\Gamma_g = \frac{Z_g - Z_0}{Z_g + Z_0}$$

Reflection coefficient at the generator

Transients On T-Lines

What happens when both the source and load impedance are mismatched?

Time for another simulation!

Code:

```
n 5 5
n 15 5
n 35 5
n 42 5
g 5 15
g 15 15
g 35 15
g 42 15
v 3:-1 p:5000:20:60 4 0
r 50 0 1
z 50 25 1 2 5 6
s 2 3
r 25 7 3
```

Transients On T-Lines





After reflecting at the load, our backwards-traveling hits the source and reflects again!



$$V_2^+ = \Gamma_g V_1^- = \Gamma_g \Gamma_L V_1^+ \qquad \Gamma_g = \frac{Z_g - Z_0}{Z_g + Z_0}$$

Transients On T-Lines





… then the reflected wave from the source (forward-moving) reflects again at the load!



$$V_2^- = \Gamma_L V_2^+ = \Gamma_L \Gamma_g V_1^- = \Gamma_g (\Gamma_L)^2 V_1^+ \qquad \Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

Transients On T-Lines

In the case of the t-line with an **impedance mismatch at both the source and the load**, how long should the circuit take (in principle) to reach steady state?

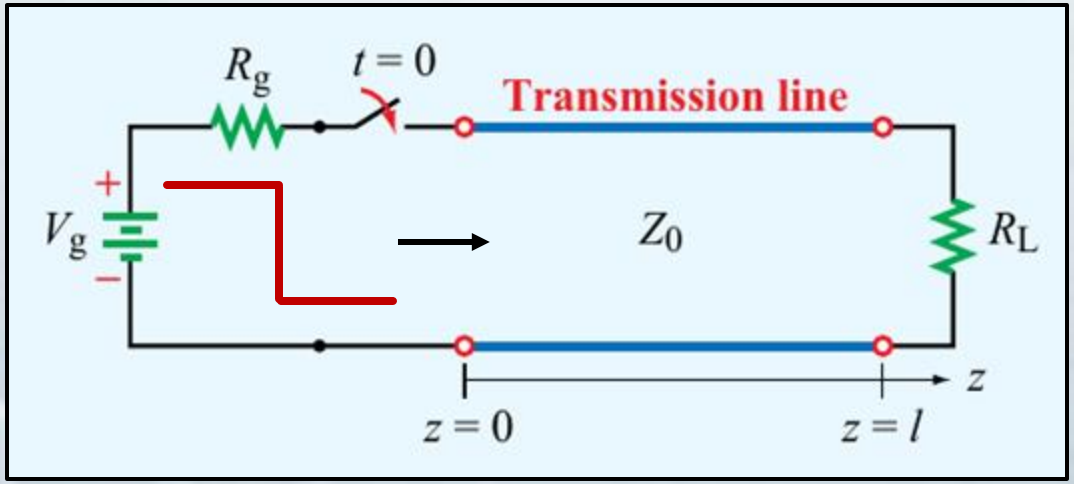


- In principle the pulse will continue bouncing between the source and the load without limit.
- In practice the pulse will fall within the measurement error of a steady-state solution after a finite number of reflections.

Transients On T-Lines
DC Source (on for t > 0)

- Now that we know what happens to the signal at the load and source impedances, let's look at DC circuit theory
- If we turn on a DC voltage source at $t = 0$ and allow the circuit to reach steady state ($t \to \infty$), what is the voltage drop across:
  - The generator/source impedance R_g
  - The load impedance R_L
  - The transmission line

Ulaby pg. 115



Transients On T-Lines

DC Source (on for t > 0)

$$
\begin{aligned}
V_\infty &= V_1^+ + V_1^- + V_2^+ + V_2^- + V_3^+ + V_3^- + \cdots \\
&= V_1^+[1 + \Gamma_L + \Gamma_L\Gamma_g + \Gamma_L^2\Gamma_g + \Gamma_L^2\Gamma_g^2 + \Gamma_L^3\Gamma_g^2 + \cdots] \\
&= V_1^+[(1+\Gamma_L)(1 + \Gamma_L\Gamma_g + \Gamma_L^2\Gamma_g^2 + \cdots)] \\
&= V_1^+(1+\Gamma_L)[1 + x + x^2 + \cdots],
\end{aligned}
$$

Ulaby pg. 117

V_∞ is the steady-state voltage or voltage reached after $t \to \infty$

$$x = \Gamma_L\Gamma_g \leqslant 1$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \ldots$$

(for |x| < 1)

Transients On T-Lines
DC Source (on for t > 0)

- So even if our pulse theoretically bounces around forever, it *does* converge to a voltage, and that voltage is:

$$V_\infty = V_1^+ \frac{1+\Gamma_L}{1-\Gamma_L\Gamma_g}$$ From converged series result

- Using our earlier equations:

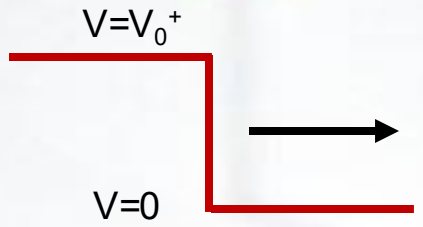
$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} \qquad \Gamma_g = \frac{Z_g - Z_0}{Z_g + Z_0} \qquad V_1^+ = \frac{V_g Z_0}{Z_g + Z_0}$$

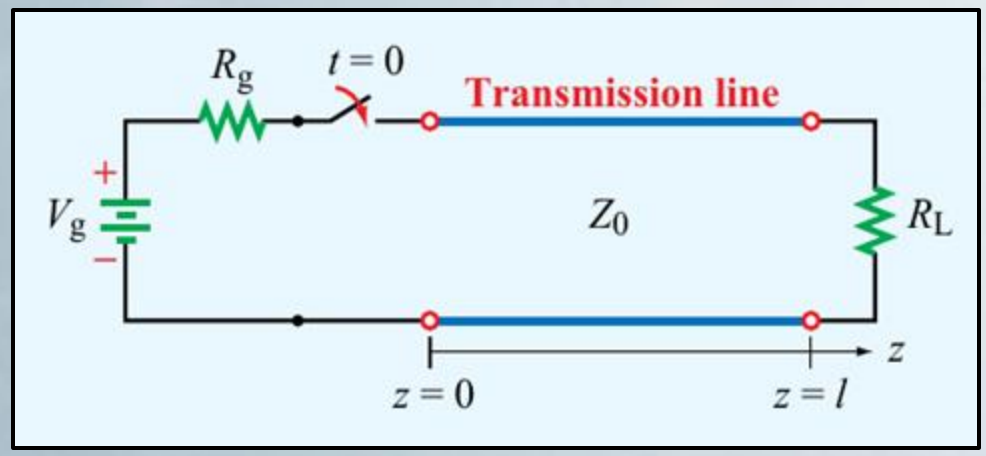
- We can write:

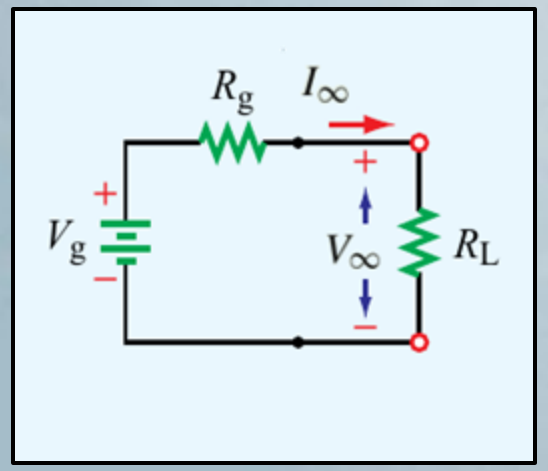
$$V_\infty = \frac{V_g R_L}{R_g + R_L}$$ = DC circuit theory

Transients On T-Lines

DC Source (on for t > 0)



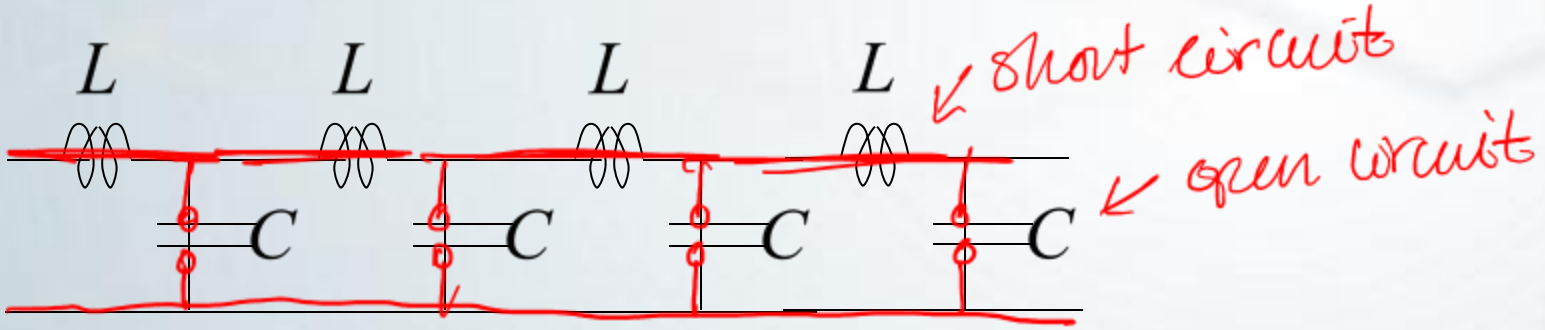




- As time increases, this t-line approaches a state in which it effectively doesn't exist anymore.
- The voltage that we put on the line is DC (doesn't change after t=0). Why don't we see this same steady-state behavior when using a sinusoidal source voltage?

Transients On T-Lines

DC Source (on for t > 0)



Keep the lumped model of the T-line in mind!

Apply a DC voltage to this line:

- What happens to the inductor impedance?
- What happens to the capacitor impedance?
- What does the lumped element model then simplify to?

For a sinusoidal input voltage, an excitation is continually applied to the inductors and capacitors → **they never reach steady state**.

Transients On T-Lines

Some things in common between transient and steady state analysis:

- Reflection at either end of the line is governed by the reflection coefficient.
- Both forward- and backward-traveling waves can be present in general.
- Z_0 is needed to understand the relationship between current and voltage on the line.

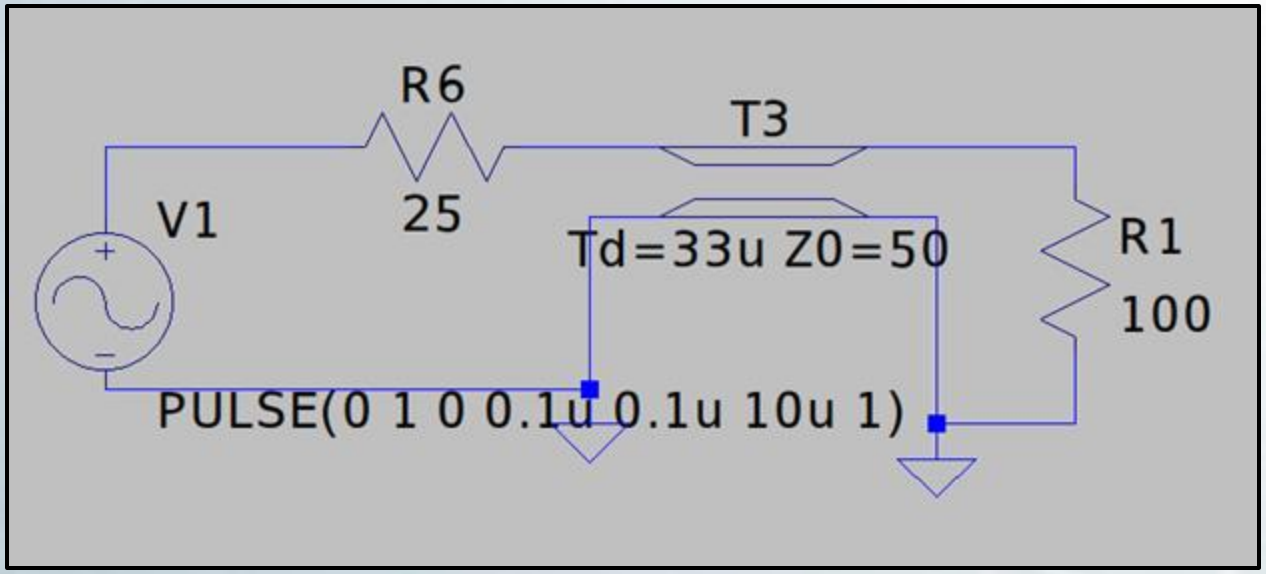
A major difference between transient and steady state analysis:

- In transient analysis, there is no backwards-traveling wave **until enough time has passed for a reflection to occur at the end of the line**.

Without forward- and backward-traveling waves on the line at the same time, a lot of our steady-state concepts (standing wave ratio, phase-shifted reflection coefficient) become meaningless.

Transients On T-Lines

Example - Square pulse, mismatched source and load



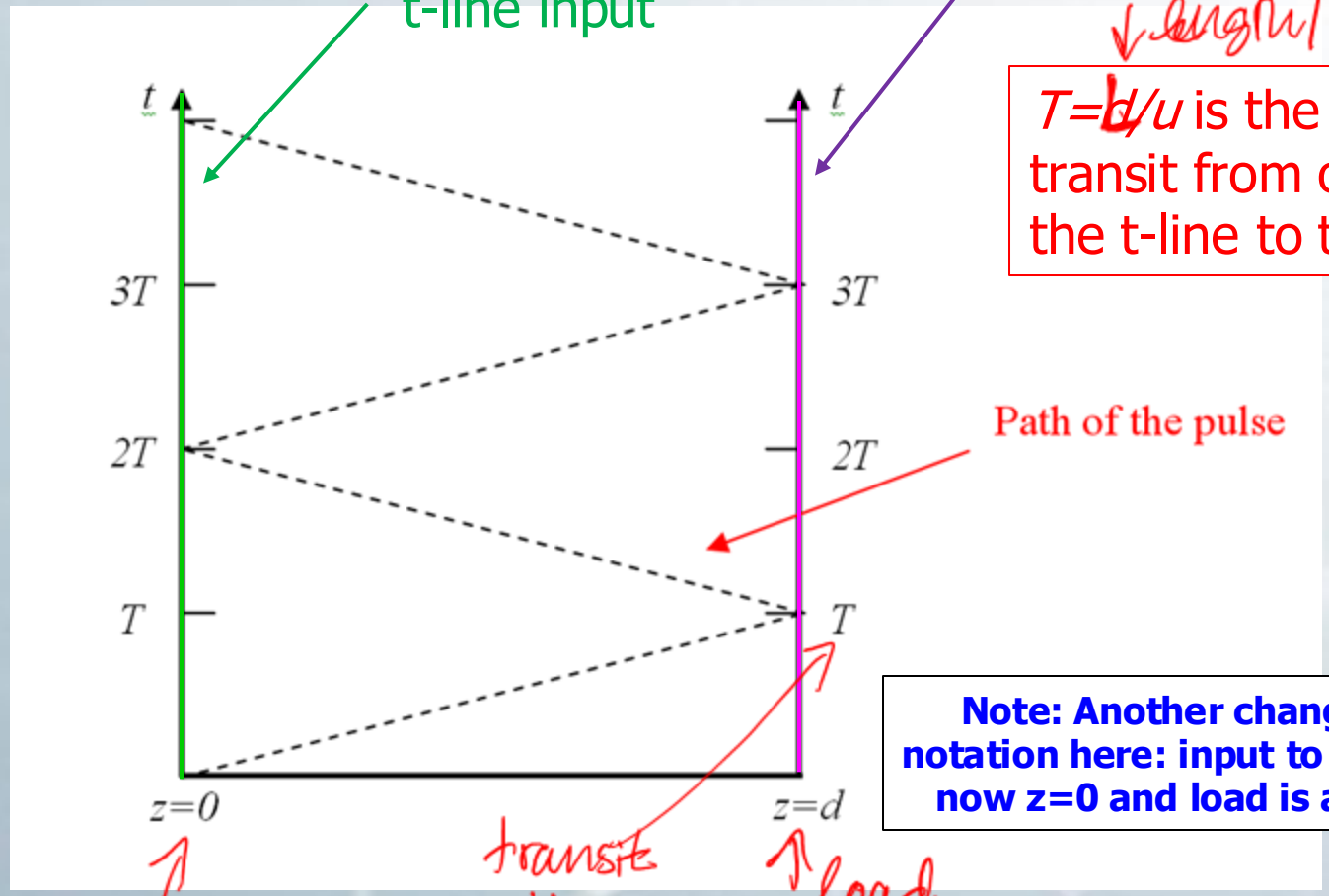
- There is a systematic method for applying this information using what is called a **bounce diagram** or **lattice diagram**
- Space and time information are both included in the diagram

Bounce Diagrams

Bounce Diagram

Scope connected to t-line input

Scope connected to t-line output

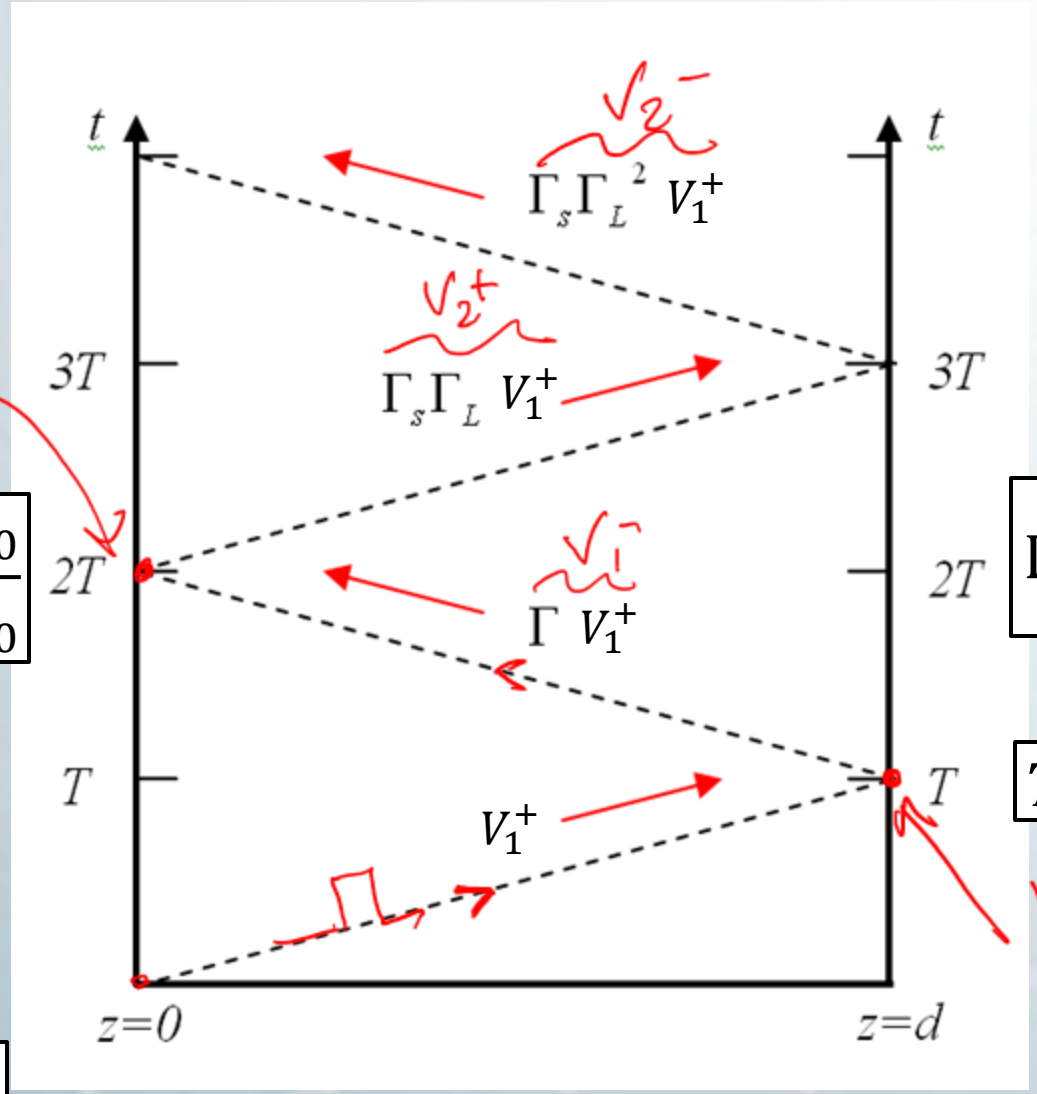


length/velocity

$T = d/u$ is the time to transit from one end of the t-line to the other

Note: Another change of notation here: input to t-line is now z=0 and load is at z=d

Bounce Diagram



$$\Gamma_g = \frac{Z_g - Z_0}{Z_g + Z_0}$$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$T = d/u$$

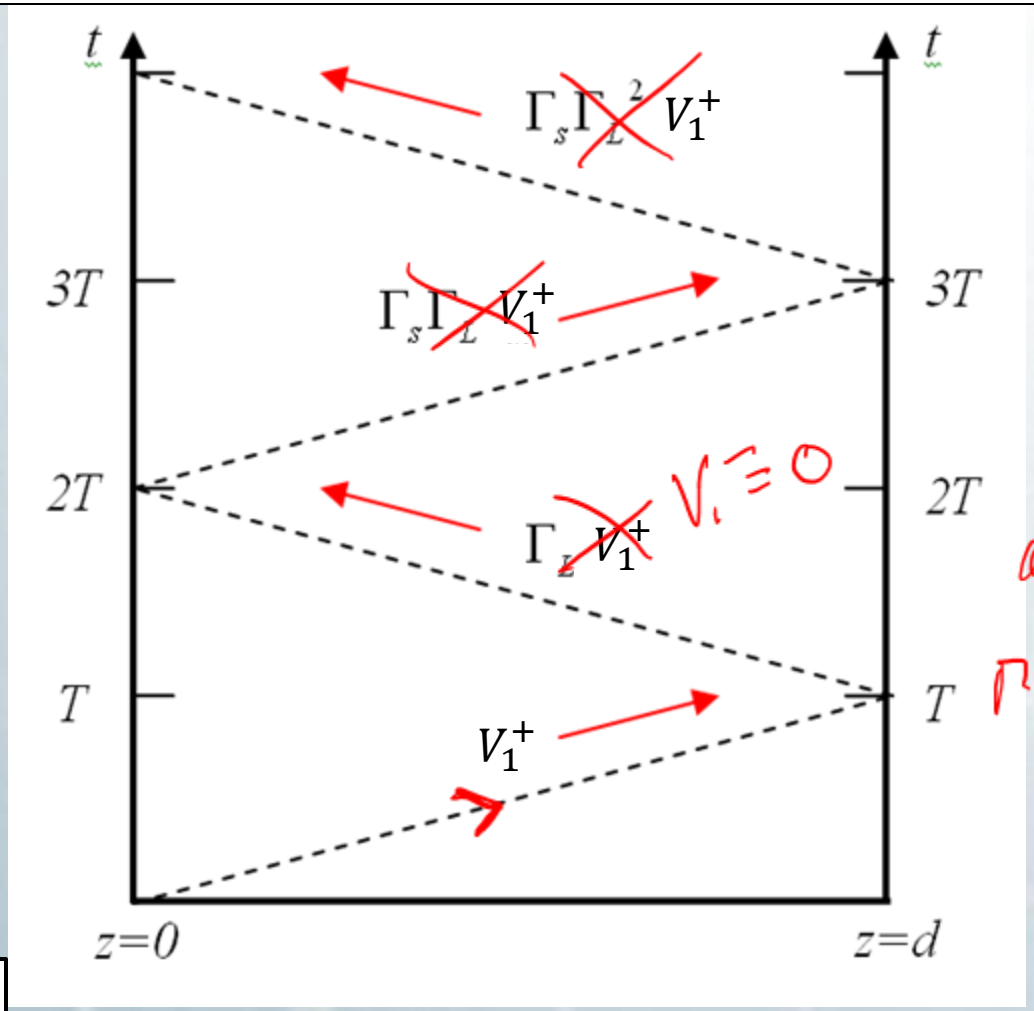
Source side

Load side

$$V_1^+ = V_g \frac{Z_0}{Z_g + Z_0}$$

Bounce Diagram

What about when the source and load impedances are matched?



all of V_1^+ transmitted to load

$\Gamma_L = 0$

Source side **Load side**

$$V_1^+ = V_g \frac{Z_0}{Z_g + Z_0}$$

Bounce Diagram

- If the pulse width is much less than the transit time T, then only a single incident and reflected pulse will be present at the load or source at a given time
- This is much simpler to consider.

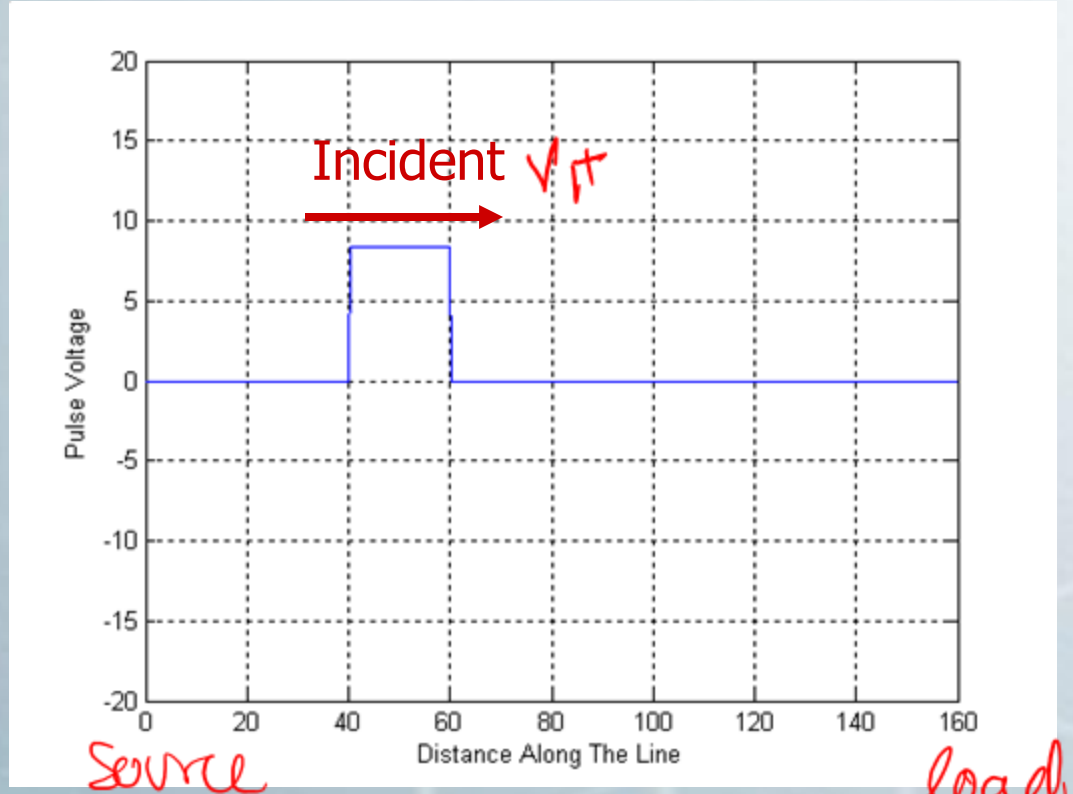




$t = 0$ $t = T_d$

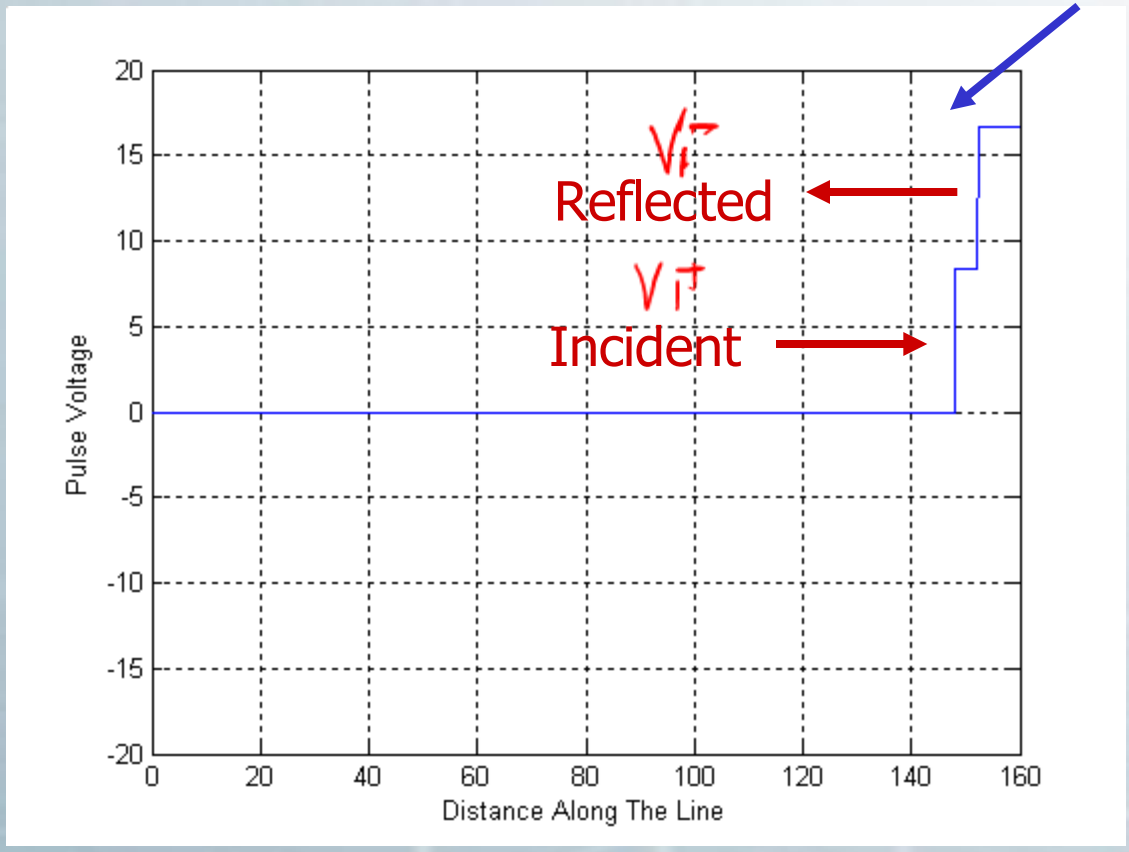
Bounce Diagram

- Zooming in allows us to see how reflection occurs in detail

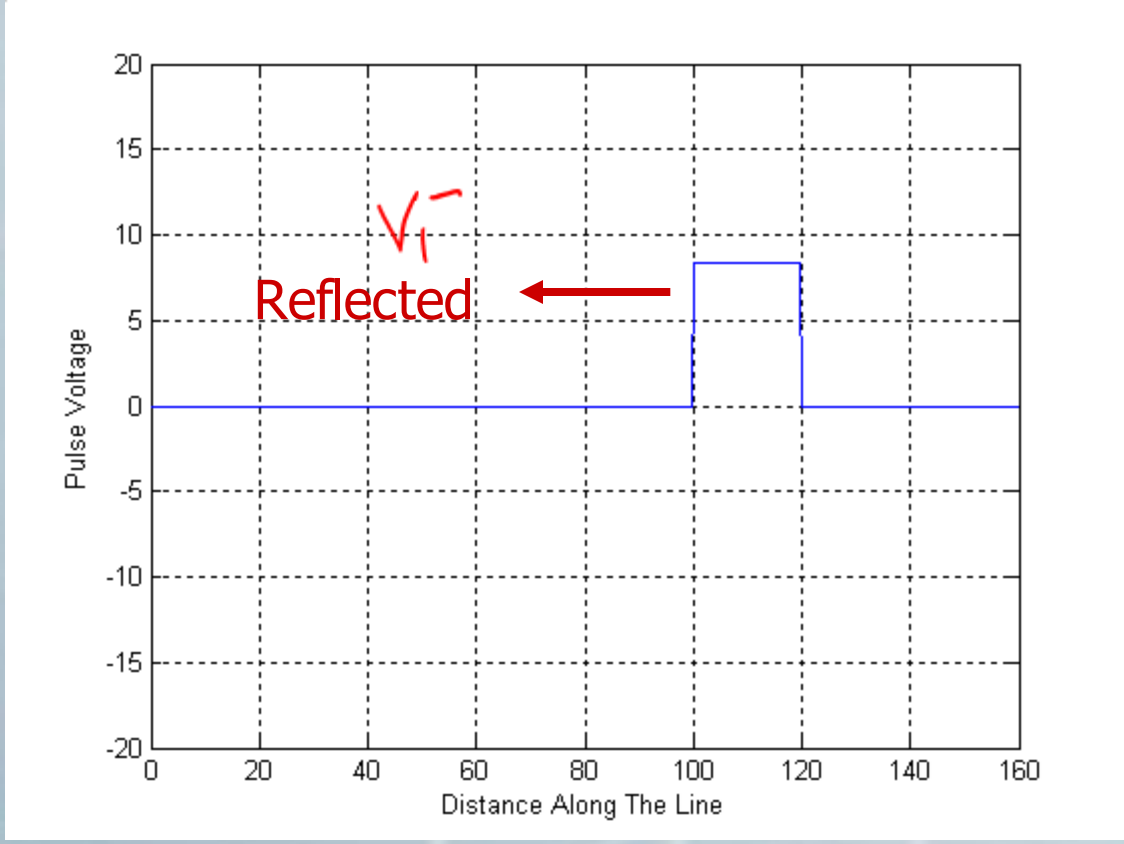


Bounce Diagram

As soon as the front edge of the incident pulse reaches the load (i.e. at $t = T_d$), it is reflected and begins moving backwards, added to the incident pulse where they overlap in space

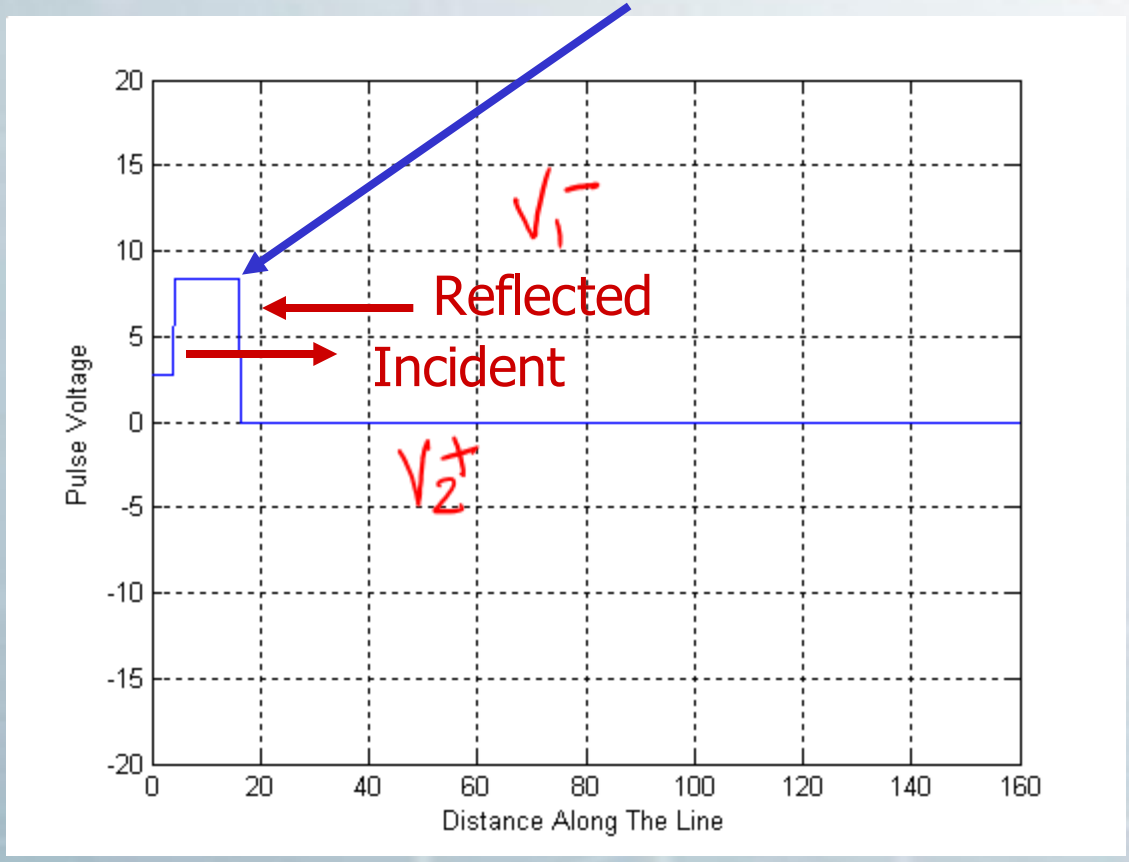


Bounce Diagram

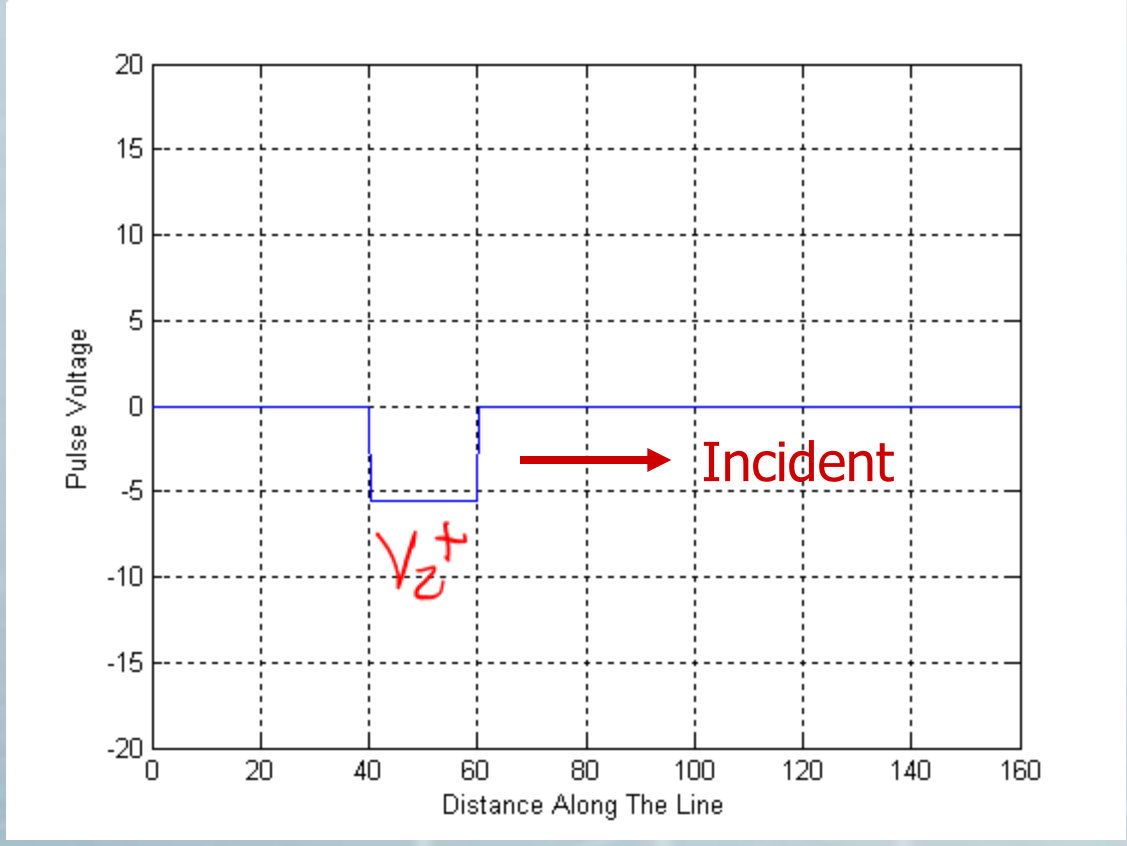


Bounce Diagram

As soon as the front edge of the reflected pulse reaches the source, it is reflected and begins moving forwards again and is added to the reflected pulse where they overlap in space

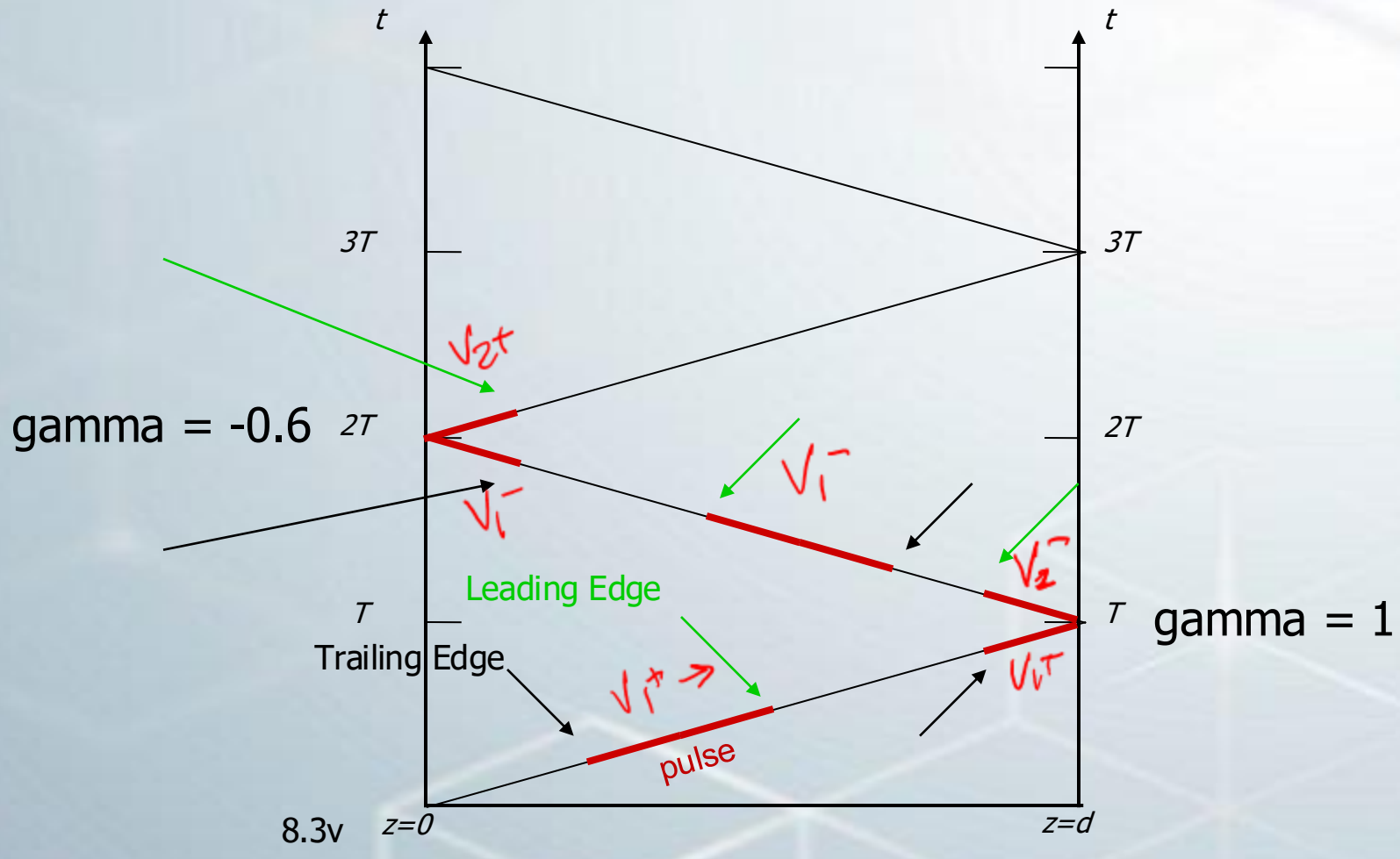


Bounce Diagram

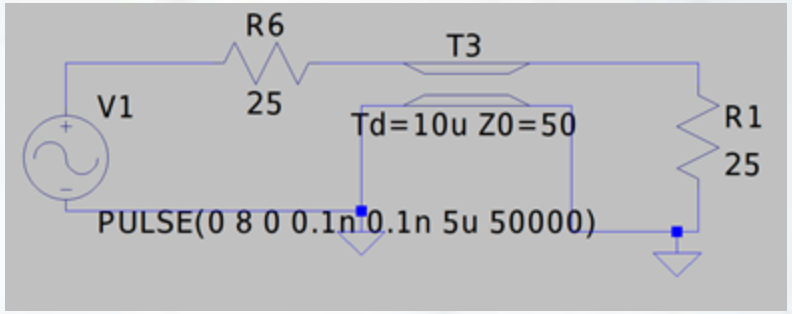


Bounce Diagram

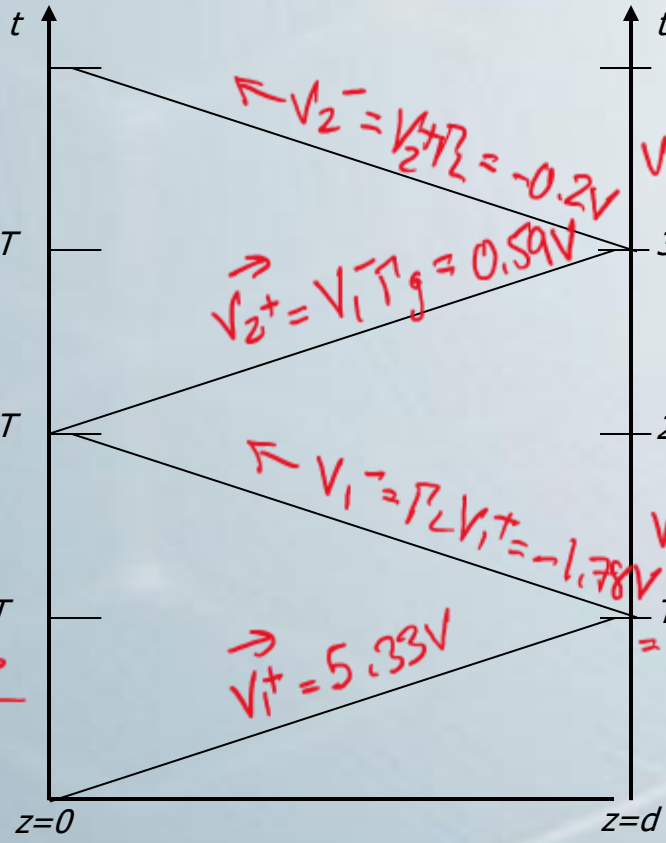
Again, expanding the pulse for clarity, we see that incident and reflected pulses exist simultaneously



Bounce Diagram Example 1 (Short Pulse)



- 8V short input pulse ($5\mu s$)
- 25Ω source and load impedance
- 50Ω characteristic impedance
- 10 microsecond T-line time delay



$V_2^- = V_{2T}^+ = -0.2V$

$V_1(3T) = V_2^+ + V_2^- = 0.39V$

$V_2^+ = V_1^- \Gamma_g = 0.59V$

$V_g(2T) = V_1^- + V_2^+ = -1.19V$

$\frac{-1.19V}{20\mu s} = 2T$

$V_1^- = \Gamma_L V_1^+ = -1.78V$

$V_L(T) = V_1^+ + V_1^- = 3.55V$

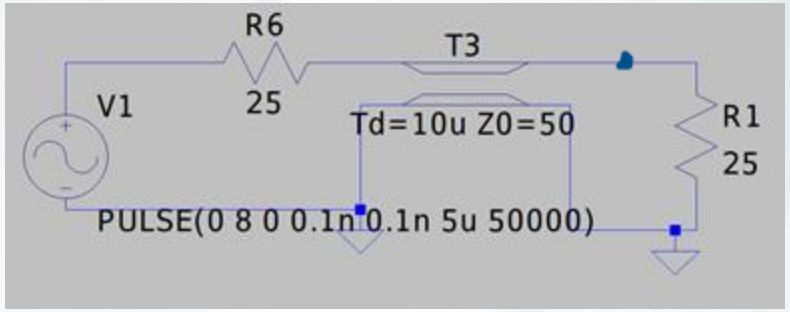
$\Gamma_g = \frac{Z_g - Z_0}{Z_g + Z_0} = -1/3$

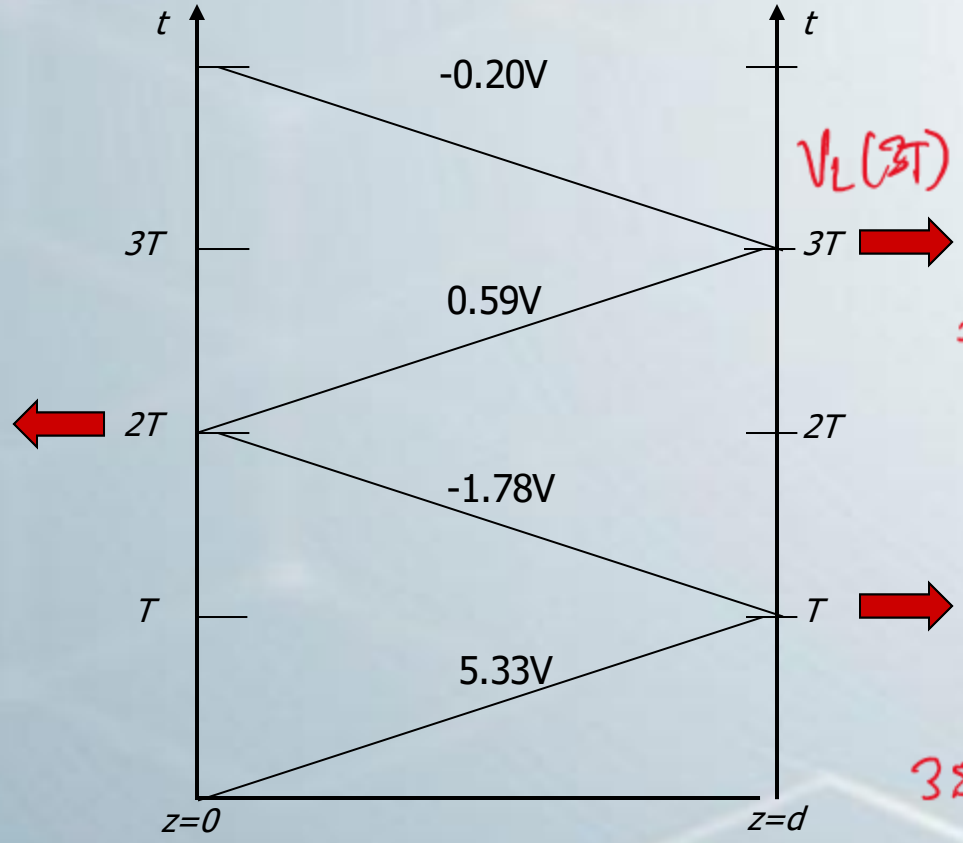
$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = -1/3$

$V_1^+ = 5.33V$

$V_1^+ = V_g \frac{Z_0}{Z_g + Z_0} = 8V \frac{50\Omega}{25\Omega + 50\Omega} = 5.33V$

Bounce Diagram Example 1 (DC Input)





$V_L(3T) = V_1^+ + V_1^- + V_2^+ + V_2^-$

$= 3.94V$

- **8V switches on at t=0 (remains on)**
- 25Ω source and load impedance
- 50Ω characteristic impedance
- 10 microsecond time delay
- Γ = -0.3333 (source and load)

$V_L(T) = V_1^+ + V_1^- = 3.55V$



$V_g(2T) = V_1^+ + V_1^- + V_2^+$

$= 4.14V$

Do Lecture 4 Exercise on Gradescope. You may work in groups of up to 4.

Wrap-Up

- Full schedule of office hours is posted on the calendar
- Studio Session 1 due tomorrow